

SYLLABUS											
Semester – 5 th											
Course Code	CT501										
Course Name	Artificial Intelligence										
Lecture (per week)	3										
Tutorial (per week)	0										
Contact Hours (per week)	3										
Total Contact Hours	36										
Credit	3										
Prerequisites: Data Structure, Design and Analysis of Algorithms, Statistics Course Objectives: The objectives of this course are to enable students to 1. Comprehend the fundamental concepts of Knowledge Representation and Inferencing in Artificial Intelligence and its utilitarian importance in current technological context 2. Formulate a problem as State-Space Exploration Framework or an Inferencing Framework of Artificial Intelligence. 3. Use the strategies of AI-Heuristics to find acceptable solutions avoiding brute-force techniques. 4. Design AI-Frameworks for Inferencing based on knowledge base. 5. Analyze the effectiveness of AI-Inferencing Model in offering solutions to the respective problem.											
Course Outcome: After successful completion of this course, students will be able to: <table border="1"> <tr> <td>CO1</td><td>Understand and explain the fundamental concepts of Knowledge Representation and Inferencing in Artificial Intelligence and its utilitarian importance in current technological context for further exploration leading towards lifelong learning.</td></tr> <tr> <td>CO2</td><td>Identify and formulate an engineering problem primarily to fit a State-Space Exploration Frame work or an Inferencing Model/Agent Design Frame work within the scope of Artificial Intelligence paradigm.</td></tr> <tr> <td>CO3</td><td>Explore relevant literature and apply the concept of Heuristic Techniques of Artificial Intelligence to solve problems.</td></tr> <tr> <td>CO4</td><td>Develop Inferencing Models for proposing solutions to the problems of Artificial Intelligence.</td></tr> <tr> <td>CO5</td><td>Implement Inferencing Models of Artificial Intelligence through developing feasible algorithms and investigate their effectiveness by analyzing their performances in solving the relevant problems.</td></tr> </table>		CO1	Understand and explain the fundamental concepts of Knowledge Representation and Inferencing in Artificial Intelligence and its utilitarian importance in current technological context for further exploration leading towards lifelong learning.	CO2	Identify and formulate an engineering problem primarily to fit a State-Space Exploration Frame work or an Inferencing Model/Agent Design Frame work within the scope of Artificial Intelligence paradigm.	CO3	Explore relevant literature and apply the concept of Heuristic Techniques of Artificial Intelligence to solve problems.	CO4	Develop Inferencing Models for proposing solutions to the problems of Artificial Intelligence.	CO5	Implement Inferencing Models of Artificial Intelligence through developing feasible algorithms and investigate their effectiveness by analyzing their performances in solving the relevant problems.
CO1	Understand and explain the fundamental concepts of Knowledge Representation and Inferencing in Artificial Intelligence and its utilitarian importance in current technological context for further exploration leading towards lifelong learning.										
CO2	Identify and formulate an engineering problem primarily to fit a State-Space Exploration Frame work or an Inferencing Model/Agent Design Frame work within the scope of Artificial Intelligence paradigm.										
CO3	Explore relevant literature and apply the concept of Heuristic Techniques of Artificial Intelligence to solve problems.										
CO4	Develop Inferencing Models for proposing solutions to the problems of Artificial Intelligence.										
CO5	Implement Inferencing Models of Artificial Intelligence through developing feasible algorithms and investigate their effectiveness by analyzing their performances in solving the relevant problems.										
MODULE NUMBER	COURSE CONTENT										
1	Introduction to Artificial Intelligence [1L] Basic Concepts, History of Artificial Intelligence, Architecture of an Artificial Intelligent Agent, Applications of Artificial Intelligence										
2	Artificial Intelligence Problem Formulation as State-Space Exploration Problem for Goal Searching [5L]										

	Basic Concepts, State-Space Exploration Formulation for Water Jug Problem, Missionaries and Cannibals Problems, Farmer-Wolf-Goat-Cabbage Problem, 8-Puzzle Problem, Constraint Satisfaction Problem and Production System for Goal Searching. Blind Search Techniques for Goal Searching: Breadth First Search, Depth First Search, Depth Limited Search, Iterative Deepening Search, Uniform Cost Search, Bi-directional Search.
3	Heuristic Techniques for Goal Searching [8L] Basic Concepts of Heuristic Techniques and Properties of Heuristic Functions, Hill Climbing Search. Best First Search, A* Search, Memory-bounded heuristic search: Iterative-deepening A* Search, Recursive Best First Search, Simplified Memory Bounded A* Search. Simulated Annealing Based Stochastic Search, Genetic Algorithm Based Evolutionary Search, Ant Colony Optimization, Particle Swarm Optimization.
4	Adversarial Search for Game Playing [2L] Basic Concepts, Minimax Search, Alpha-Beta Pruning.
5	Knowledge Representation and Inference using Propositional Logic and Predicate Logic[5L] Propositional Logic: Knowledge Representation and Inference using Propositional Logic Predicate Logic: Knowledge Representation, Inference and Answer Extraction using First Order Predicate Logic
6	Slot-and-Filler Structure for Knowledge Representation [2L] Weak Slot-and-Filler Structure for Knowledge Representation: Semantic Nets and Frames. Strong Slot-and-Filler Structure for Knowledge Representation: Conceptual Dependency and Script.
7	Reasoning under Uncertainty [5L] Bayesian Inferencing and Bayesian Belief Network, Dempster-Shafer Theory, Overview of Fuzzy Logic and Inferencing, Overview of Hidden Markov Model
8	Planning [5L] Basic Concepts, Problem of Blocks World, Components of a Planning System, Algorithms for Planning: Goal Stack, Nonlinear Planning Using Constraint Posting, Hierarchical Planning, Algorithms for Planning as State-Space Search, Heuristics for planning, Planning Graphs and GRAPHPLAN Algorithm.
9	Introduction to Natural Language Processing [1L] Basic Concepts, Steps of Natural Language Processing, Morphological, Syntactic and Semantic Analysis, Discourse Integration and Pragmatic Analysis, Applications of Natural Language Processing.
10	Introduction to Machine Learning [2L]

	Basic concepts of Machine Learning Model, Supervised Learning, Unsupervised Learning, and Reinforced Learning, Overview of Artificial Neural Network
Textbook: 1. Russell, S. and Norvig, P. 2015. Artificial Intelligence - A Modern Approach, 3rd edition, Prentice Hall. 2. Rich, E., Knight, K and Shankar, B. 2009. Artificial Intelligence, 3rd edition, Tata McGrawHill. Reference Books: 1. Padhy, N.P. 2009. Artificial Intelligence and Intelligent Systems, Oxford University Press. 2. Deepak Khemani, “A First Course in Artificial Intelligence”, McGraw Hill.	

CO-PO Mapping

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	2	-	-	-	-	-	-	-	-	-	3	3	-	-
CO2	2	3	-	-	-	-	-	-	-	-	-	-	-	3	-
CO3	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
CO4	2	2	2	3	-	-	-	-	-	-	-	2	2	-	2
CO5	2	2	3	3	2				-	-	-	2	2	2	3

SYLLABUS	
Semester – 5 th	
Course Code	CT502
Course Name	Database Management Systems
Lecture (per week)	3
Tutorial (per week)	0
Contact Hours (per week)	3
Total Contact Hours	36
Credit	3
Prerequisites: 1. Logic of programming language 2. Basic concepts of data structure and algorithms	
Course Objectives: 1. To Understand and Describe the basic concepts and utility of Database management system, different data models of Database management system. 2. To Design an Entity Relationship (E-R) Diagram and relational model for any kind of real-life application and able to Apply relational algebra operations, SQL, Neo4j for solving query. 3. To Analyze and Create the relational database for any real-life applications based on normalization. 4. To Apply the query optimization techniques, different file organization techniques and determine whether the transaction satisfies the ACID properties.	
Course Outcome: After successful completion of this course, students will be able to:	
CO1	To Understand and Describe the basic concepts and utility of Database management system, different data models of Database management system.
CO2	To Design an Entity Relationship (E-R) Diagram and relational model for any kind of real-life application and able to Apply relational algebra operations, SQL, Neo4j for solving query.
CO3	To Analyze and Create the relational database for any real-life applications based on normalization.
CO4	To Apply the query optimization techniques, different file organization techniques and determine whether the transaction satisfies the ACID properties.
CO5	Explore DBMS based ideas through developing software programs with adequate documentation in collaborative environment for successfully carrying out projects on DBMS Problems and investigate their effectiveness by analyzing the performances using proper techniques and tools and Assess the limitations of solutions underscoring utilitarian importance for further explorations leading towards lifelong learning.
MODULE NUMBER	COURSE CONTENT

[illegible]

C03	2	2	3	2	-	-	-	-	-	-	-	-	2	-	2
C04	2	2	2	3	-	-	-	-	-	-	-	-	2	-	2
C05	2	2	3	3	-	-	-	-	-	-	-	2	2	2	3

SYLLABUS	
Semester – 5 th	
Course Code	CT503
Course Name	Object Oriented Programming using Java
Lecture (per week)	3
Tutorial (per week)	1
Contact Hours (per week)	4
Total Contact Hours	48
Credit	4
Prerequisites: - Computer Fundamentals - Basic understanding of Computer Programming and related Programming Paradigms Problem Solving Techniques with proper logic Implementation	
Course Objectives: The objectives of this course are to enable students to - It allows to map with real world Object (Object orientation) rather than action(Procedure) that comes to produce software as separated code modules which rise up decoupling and increases code re-usability. - It demonstrates that how can you change the implementation of an object without affecting any other code by increasing data security and protecting unwanted data access. (Encapsulation). - It allows you to have many different functions, all with the same name, all doing the same job, but depending upon different data. (Polymorphism). - It guides you to write generic code: which will work with a range of data, so you don't have to write basic stuff over, and over again. (Generics). - It lets you write a set of functions, then expand them in different direction without changing or copying them in any way. (Inheritance)	
Course Outcome: After successful completion of this course, students will be able to:	
CO1	Design the process of interaction between Objects, classes & methods w.r.t. Object Oriented Programming.
CO2	Acquire a basic knowledge of Object Orientation with different properties as well as different features of Java.
CO3	Analyze various activities of different string handling functions with various I/O operations.
CO4	Discuss basic code reusability feature w.r.t. Inheritance, Package and Interface.
CO5	Implement Exception handling, Multithreading and Applet (Web program in java) programming concept in Java.
MODULE NUMBER	COURSE CONTENT

1	Introduction [4L] Object Oriented Analysis (OOA) & Object Oriented Design (OOD) - Concepts of object oriented programming language, Relationships among objects and classes-Generalization, Specialization, Aggregation, Association, Composition, links, Meta-class. [2L]; Object Oriented Programming concepts - Difference between Java and C++; Different features of Java [2L];
2	Java Basics [12L] Basic concepts of java programming - Advantages of java, Byte-code & JVM, Data types, Different types of Variables.[2L] ;Java Operators & Control statements [1L]; Java loops. [1L]; Array.[1L] ;Creation of class, object, method. [2L]; Constructor- Definition, Usage of Constructor, Different types of Constructor.[1L]; finalize method and garbage collection, Method & Constructor overloading. [1L]; this keyword, use of objects as parameter & methods returning objects.[1L]; Call by value & call by reference. [1L]; Static variables & methods. Nested & inner classes.[1L].
3	Basic String handling & I/O [8L] Basic string handling concepts- Concept of mutable and immutable string, Methods of String class- charAt(), compareTo(), equals(), equalsIgnoreCase(), indexOf(), length() , substring(). [2L]; toCharArray(), toLowerCase(), toString(), toUpperCase() , trim() , valueOf() methods, Methods of String buffer class- append(), capacity(), charAt(), delete(), deleteCharAt(). [2L];ensureCapacity(), getChars(), indexOf(), insert(), length(), setCharAt(), setLength(), substring(), toString(). [2L] ;Command line arguments, basics of I/O operations – keyboard input using BufferedReader[1L] ; Scanner class in Java I/O operation [1L];
4	Inheritance and Java Package[10L] Inheritance - Definition, Advantages, Different types of inheritance and their implementation. [1L] ;Super and final keywords, super() method. [2L]; Method overriding, Dynamic method dispatch.[1L]; Abstract classes & methods.[2L]; Interface - Definition, Use of Interface.[1L]; Multiple inheritance by using Interface.[1L] ;Java Packages -Definition, Creation of packages. [1L]; Java Access Modifiers - public, private, default and protected,Importing packages, member access for packages. [1L]
5	Exception handling, Multithreading and Applet Programming [14L] Exception handling - Basics, different types of exception classes. Difference between Checked & Unchecked Exception.[2L]; Try & catch related case studies.[1L]; Throw, throws & finally. [2L]; Creation of user defined exception. [1L]; Multithreading - Basics, main thread [1L]; Thread life cycle.[1L]; Creation of multiple threads-yield(), suspend(), sleep(n), resume(), wait(), notify(), join(), isAlive().[1L] ;Thread priorities, thread synchronization.[1L]; Interthread communication, deadlocks for threads[1L]; Applet Programming - Basics, applet life cycle, difference between application & applet programming[2L]; Parameter passing in applets. [1L]
Textbook: 1. Herbert Schildt – "Java: The Complete Reference " – 9 th Ed. – TMH 2. E. Balagurusamy – " Programming With Java: A Primer " – 3rd Ed. – TMH. Reference Books: 1. R.K Das – " Core Java for Beginners " – VIKAS PUBLISHING. 2. Rumbaugh, James Michael, Blaha – " Object Oriented Modelling and Design " – Prentice Hall, India.	

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	2	3	3	2	-	-	-	-	-	-	-	-	-	2	3
CO2	3	2	3	1	-	-	-	-	-	-	-	-	3	2	-
CO3	3	3	2	3	-	-	2	-	1	-	-	-	2	-	3
CO4	2	-	2	2	-	-	-	-	-	-	-	-	3	2	-
CO5	2	-	3	1	2	-	-	-	2	-	2	-	-	2	3

SYLLABUS											
Semester – 5 th											
Course Code	CT504A										
Course Name	Computer Graphics										
Lecture (per week)	3										
Tutorial (per week)	0										
Contact Hours (per week)	3										
Total Contact Hours	36										
Credit	3										
Prerequisites: Mathematics, Computer Fundamentals & Principle of Computer Programming Course Objectives: 1. The objectives of this course are to enable students to Use of the component so fa graphics system and become familiar with building approach of graphics system components and algorithms related with them. 2. Understand thebasicprinciplesof2Dand3Dcomputergraphics. 3. Understand of how to scan convert the basic geometrical primitives, how to transform the shapes to fit the master the picture definition. 4. Understand the mapping from a world co-ordinates to device co-ordinates, clipping, and projections . 5. Discuss the application of computer graphics concepts in the development of computer games, information visualization, and business applications.											
Course Outcome: After successful completion of this course, students will be able to: <table border="1"> <tr> <td>CO1</td><td>Understand the fundamental concept of Computer graphics and mathematical knowledge and explain the foundations of computer graphics and different display technology and devices.</td></tr> <tr> <td>CO2</td><td>Demonstrate different scan conversion algorithms, drawing algorithms, polygon filling algorithms, curves and surface drawing algorithms, clipping algorithms, surface removal algorithms using graphics tools.</td></tr> <tr> <td>CO3</td><td>Understand the basic concept of graphics programming and implementclippingwiththecomprehensionofwindows,view-portsinrelationtoimages display on screen.</td></tr> <tr> <td>CO4</td><td>Analyze and compare different drawing algorithms, polygon filling algorithms, curves and surface drawing algorithms hidden surface illumination methods</td></tr> <tr> <td>CO5</td><td>Develop the concept of geometric models, mathematical and algorithmic approach necessary for programming computer graphics leading to lifelong learning.</td></tr> </table>		CO1	Understand the fundamental concept of Computer graphics and mathematical knowledge and explain the foundations of computer graphics and different display technology and devices.	CO2	Demonstrate different scan conversion algorithms, drawing algorithms, polygon filling algorithms, curves and surface drawing algorithms, clipping algorithms, surface removal algorithms using graphics tools.	CO3	Understand the basic concept of graphics programming and implementclippingwiththecomprehensionofwindows,view-portsinrelationtoimages display on screen.	CO4	Analyze and compare different drawing algorithms, polygon filling algorithms, curves and surface drawing algorithms hidden surface illumination methods	CO5	Develop the concept of geometric models, mathematical and algorithmic approach necessary for programming computer graphics leading to lifelong learning.
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CO5	Develop the concept of geometric models, mathematical and algorithmic approach necessary for programming computer graphics leading to lifelong learning.										

MODULE NUMBER	COURSE CONTENT
1	Introduction [4L] Introduction: Objective, applications, GKS/PHIGS, normalized co-ordinate system, aspect ratio.
2	Computer Graphics System [4L] Graphics System: Vector and raster graphics, various graphics display devices, graphics interactive devices, segmented graphics, attribute table.
3	Computer Graphics System [4L] Raster Scan Graphics: Line drawing algorithms, circle/ellipse drawing algorithms, polygon filling algorithms.
4	Geometric Transformation [4L] Geometric Transformation: Homogeneous co-ordinate system, 2D and 3D transformations, projection— orthographic and perspective.
5	Curves and Surfaces [4L] Curves and Surfaces: Curve approximation and interpolation, Lagrange, Hermite, Bezier and B Spline curves/surfaces and their properties, curves and surface drawing algorithms.
6	Curves and Surfaces 2 [4L] Geometric modeling: 3D object representation and its criteria, edge/vertex list, constructive solid geometry, wire-frame model, generalized cylinder, finite element methods.
7	Viewing and Clipping [4L] Clipping: Window and viewport, 2D and 3D clipping algorithms.
8	Hidden Surfaces [4L] Hidden Lines and Hidden Surfaces: Concept of object- and image-space methods, lines and surface removal algorithms.
9	Illumination and Color models [4L] Intensify, Coloring and Rendering: RGB, YIQ, HLS and HSV models and their conversions, gamma correction, half toning. Illumination models, polygon mesh shading, transparency, shadow, texture.

Textbook:

1. D. Hearn and P. M. Baker: Computer Graphics, 2nd ed. Prentice Hall of India, New Delhi, 1997.
2. W. M. Newman and R. F. Sproull: Principles of Interactive Computer Graphics, McGraw Hill, New Delhi, 1979.

Reference Books:

1. F. S. Hill: Computer Graphics, McMillan, New York, 1990.
2. D. P. Mukherjee: Fundamentals of Computer Graphics and Multimedia, Prentice Hall of India, New Delhi, 1999.
3. J. D. Foley et al.: Computer Graphics, 2nd ed., Addison-Wesley, Reading, Mass., 1993.

[illegible]

SYLLABUS	
Semester – 5 th	
Course Code	CT504B
Course Name	Network Security and Cryptography
Lecture (per week)	3
Tutorial (per week)	0
Contact Hours (per week)	3
Total Contact Hours	36
Credit	3
Prerequisites: Computer Fundamentals & knowledge in mathematics Course Objectives: 1. The objective of this course is to teach the concepts of securing computer network protocols, based on the application of cryptography techniques. 2. Basic of cryptography, including conventional and public-key cryptography, hash functions, authentication, and digital signatures. 3. Key Management and Distribution: Symmetric Key Distribution, Distribution of Public Keys, X.509 Certificates, Public-Key Infrastructure. 4. User Authentication: Remote User-Authentication Principles, Remote User-Authentication Using Symmetric Encryption, Kerberos Systems, Remote User Authentication Using Asymmetric Encryption. 5. Malicious Software: Viruses, Worms, System Corruption, Attack Agents, Information Theft Key loggers, Phishing, Spyware Payload Stealthing, Backdoors, Rootkits, Distributed Denial of Service Attacks. 6. Network Access Control: Network Access Control, Extensible Authentication Protocol, IEEE 802.1X Port-Based Network Access Control. 7. IP Security: IP Security Overview, IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange (IKE).	
Course Outcome: After successful completion of this course, students will be able to:	
CO1	Understand the fundamental concept of Network security and explain the foundations of cryptography.
CO2	Demonstrate the use of basic solutions using cryptography, including conventional and public-key cryptography, hash functions, authentication, and digital signatures.
CO3	Understand the basic concept of Symmetric Key Distribution, Distribution of Public Keys, Public-Key Infrastructure.
CO4	Analyze remote User-Authentication Principles, Remote User-Authentication Using Symmetric Encryption, Kerberos Systems, Remote User Authentication Using Asymmetric Encryption.

CO5	Develop IP Security Policy, Encapsulating Security Payload, Combining Security Associations, Internet Key Exchange (IKE).	
MODULE NUMBER	COURSE CONTENT	
1	<u>Module I [6]:</u> Security Concepts: Introduction, The need for security, Security approaches, Principles of security, Types of Security attacks, Security services, Security Mechanisms, A model for Network Security Cryptography Concepts and Techniques: Introduction, plain text and cipher text, substitution techniques, transposition techniques, encryption and decryption, symmetric and asymmetric key cryptography, steganography, key range and key size, possible types of attacks	
2	<u>Module II [8]:</u> Symmetric key Ciphers: Block Cipher principles, DES, AES, Blowfish, RC5, IDEA, Block cipher operation, Stream ciphers, RC4. Asymmetric key Ciphers: Principles of public key cryptosystems, RSA algorithm, Elgamal Cryptography, Diffie-Hellman Key Exchange, Knapsack Algorithm.	
3	<u>Module III [8]:</u> Cryptographic Hash Functions: Message Authentication, Secure Hash Algorithm (SHA-512), Message authentication codes: Authentication requirements, HMAC, CMAC, Digital signatures, Elgamal Digital Signature Scheme. Key Management and Distribution: Symmetric Key Distribution Using Symmetric & Asymmetric Encryption, Distribution of Public Keys, Kerberos, X.509 Authentication Service, Public – Key Infrastructure	
4	<u>Module IV [8]:</u> Transport-level Security: Web security considerations, Secure Socket Layer and Transport Layer Security, HTTPS, Secure Shell (SSH) Wireless Network Security: Wireless Security, Mobile Device Security, IEEE 802.11 Wireless LAN, IEEE 802.11i Wireless LAN Security	
5	<u>Module V [6]:</u> E-Mail Security: Pretty Good Privacy, S/MIME IP Security: IP Security overview, IP Security architecture, Authentication Header, Encapsulating security payload, Combining security associations, Internet Key Exchange Case Studies on Cryptography and security: Secure Multiparty Calculation, Virtual Elections, Single sign On, Secure Inter-branch Payment Transactions, Cross site Scripting Vulnerability.	
Textbook:		
1. Cryptography and Network Security: Principles and Practice, 6th Edition, William Stallings, 2014, Pearson, ISBN13:9780133354690.		
Reference Books:		
1. Network Security: Private Communications in a Public World, M. Speciner, R. Perlman, C. Kaufman, Prentice Hall, 2002. 2. Linux iptables Pocket Reference, Gregor N. Purdy, O'Reilly, 2004, ISBN-13: 978-0596005696. 3. Linux Firewalls, by Michael Rash, No Starch Press, October 2007, ISBN: 978-1-59327-141-1.		

[illegible]

SYLLABUS	
Semester – 5 th	
Course Code	CT504C
Course Name	Compiler Design
Lecture (per week)	3
Tutorial (per week)	0
Contact Hours (per week)	3
Total Contact Hours	36
Credit	3
Prerequisites: - Mathematics - Concept of programming languages - Data structures - Computer architecture - Formal languages and automata theory - Some advanced math might be required if you adventure in code optimization	
Course Objectives: The objectives of this course are to enable students to - Make the student understand the process involved in a compiler - Create an overall view of various types of translators, linkers, loaders, and phases of a compiler - Understand the concepts of syntax analysis, various types of parsers especially the top down approach - Create awareness among students the various types of bottom up parsers, - Understand the syntax analysis and, intermediate code generation, type checking, the role of symbol table and its organization, Code generation, machine independent code optimization and instruction scheduling.	
Course Outcome: After successful completion of this course, students will be able to:	
CO1	Illustrate the basic concept of compilers and discuss on the components as well as the strengths and weaknesses of various phases of designing a compiler.
CO2	Design and analyze algorithms for syntactic or parsing techniques and semantic analysis of the process of designing compilers.
CO3	Develop the parsers and experiment the knowledge of activation tree, activation record and dynamic storage allocation techniques
CO4	Construct the intermediate code representations and generation.
CO5	Apply for various optimization techniques for dataflow analysis.
MODULE NUMBER	COURSE CONTENT
1	Module I [7L] Compilers, Cousins of the Compiler, Analysis-synthesis model, The phases of the compiler. The role of the lexical analyzer, Tokens, Patterns, Lexemes, Input buffering, Specifications of a token, Recognition of tokens, Finite automata, From a regular expression to an NFA, From a regular expression to DFA, Design of a lexical analyzer generator (Lex).

[illegible]

SYLLABUS	
Semester – 5 th A. THEORY	
Course Code	HU(CT)501
Course Name	Economics for Engineers
Lecture [per week]	2
Tutorial [per week]	0
Contact Hours [per week]	2
Total Contact Hours	24
Credit	2
Pre-requisites: MATH – College Algebra, Pre-Calculus Algebra and Trigonometry.	
Course Outcome[s]: On completion of the course students will be able to	
CO1	Apply the appropriate engineering economics analysis method[s] for problem solving: present worth, annual cost, rate-of-return, payback, break-even, benefit-cost ratio.
CO2	Evaluate the cost effectiveness of individual engineering projects using the methods learned and drawing references for the investment decisions.
CO3	Compare the life cycle cost of multiple projects using the methods learned, and make a quantitative decision between alternate facilities and/or systems.
CO4	Evaluate the profit of a firm, carry out the break even analysis and employ this tool to make production decision.
CO5	Discuss and solve advanced economic engineering analysis problems including taxation and inflation.
MODULE NUMBER	COURSE CONTENT
1	Introduction[3L] Managerial Economics-Relationship with other disciplines-Firms: Types, Objectives and goals-Manual Decisions-Decision Analysis
2	Demand and Supply Analysis[5L] Demand-Types of demand-determinants of demand-Demand function-Demand Elasticity-Demand forecasting-Supply-Determinants of supply-Supply function-Supply Elasticity.
3	Cost Analysis [5L] Element of costs, Marginal cost, Marginal Revenue, Sunk cost, Opportunity cost, Break-even analysis– PV ratio.
4	Elementary economic Analysis [4L] Inflation-Meaning of inflation, types, causes, measures to control inflation. National Income-Definition, Concepts of national income, Method of measuring national income.
5	Financial Accounting [5L] Concepts and Definition of Accounting, Journal, Ledger, Trial Balance. Trading A/C, Profit & Loss A/C and Balance Sheet.

[illegible]

SYLLABUS											
Semester – 5 th											
Course Code	CT591										
Course Name	Artificial Intelligence Lab										
Lecture (per week)	0										
Tutorial (per week)	0										
Contact Hours (per week)	3										
Total Contact Hours	36										
Credit	1.5										
Prerequisites: Data Structure, Design and Analysis of Algorithms, Statistics											
Course Objectives: The objectives of this course are to enable students to 1. Gain foundational knowledge of PROLOG to implement an Artificial Intelligent Agent as an executable computer program for Knowledge Representation and Inferencing 2. Formulate a problem by analyzing its characteristics to fit a State-Space Exploration Framework or an Inferencing Frame work of Artificial Intelligence. 3. Apply the concepts of Artificial Intelligence to solve a problem by implementing well-known Artificial Intelligence strategies using proper techniques and tools of PROLOG. 4. Build expert systems offering solutions to the challenging problems of Artificial Intelligence. 5. Implement Artificial Intelligence based ideas as executable PROLOG programs through developing intelligent heuristic strategies											
Course Outcome: After successful completion of this course, students will be able to: <table border="1"> <tr> <td>CO1</td><td>Acquire foundational knowledge of PROLOG to implement an Artificial Intelligent Agent as an executable computer program for Knowledge Representation and Inferencing and understand the working principle of the agent and assess its utilitarian importance in current technological context leading towards lifelong learning.</td></tr> <tr> <td>CO2</td><td>Identify and formulate an engineering problem by analyzing its characteristics to fit a State-Space Exploration Framework or an Inferencing Agent Formulation Framework of Artificial Intelligence.</td></tr> <tr> <td>CO3</td><td>Explore relevant literature and apply the concepts of Artificial Intelligence to solve a problem by implementing well-known Artificial Intelligence strategies using proper techniques and tools of PROLOG.</td></tr> <tr> <td>CO4</td><td>Develop ideas and propose expert systems offering solutions to the challenging problems of Artificial Intelligence.</td></tr> <tr> <td>CO5</td><td>Plan and Implement Artificial Intelligence base did eas as executable PROLOG programs through developing intelligent heuristic strategies or expert systems with adequate documentation in collaborative environment for successfully carrying out projects on Artificial Intelligence Problems and investigate their effectiveness by analyzing the performances using proper techniques and tools.</td></tr> </table>		CO1	Acquire foundational knowledge of PROLOG to implement an Artificial Intelligent Agent as an executable computer program for Knowledge Representation and Inferencing and understand the working principle of the agent and assess its utilitarian importance in current technological context leading towards lifelong learning.	CO2	Identify and formulate an engineering problem by analyzing its characteristics to fit a State-Space Exploration Framework or an Inferencing Agent Formulation Framework of Artificial Intelligence.	CO3	Explore relevant literature and apply the concepts of Artificial Intelligence to solve a problem by implementing well-known Artificial Intelligence strategies using proper techniques and tools of PROLOG.	CO4	Develop ideas and propose expert systems offering solutions to the challenging problems of Artificial Intelligence.	CO5	Plan and Implement Artificial Intelligence base did eas as executable PROLOG programs through developing intelligent heuristic strategies or expert systems with adequate documentation in collaborative environment for successfully carrying out projects on Artificial Intelligence Problems and investigate their effectiveness by analyzing the performances using proper techniques and tools.
CO1	Acquire foundational knowledge of PROLOG to implement an Artificial Intelligent Agent as an executable computer program for Knowledge Representation and Inferencing and understand the working principle of the agent and assess its utilitarian importance in current technological context leading towards lifelong learning.										
CO2	Identify and formulate an engineering problem by analyzing its characteristics to fit a State-Space Exploration Framework or an Inferencing Agent Formulation Framework of Artificial Intelligence.										
CO3	Explore relevant literature and apply the concepts of Artificial Intelligence to solve a problem by implementing well-known Artificial Intelligence strategies using proper techniques and tools of PROLOG.										
CO4	Develop ideas and propose expert systems offering solutions to the challenging problems of Artificial Intelligence.										
CO5	Plan and Implement Artificial Intelligence base did eas as executable PROLOG programs through developing intelligent heuristic strategies or expert systems with adequate documentation in collaborative environment for successfully carrying out projects on Artificial Intelligence Problems and investigate their effectiveness by analyzing the performances using proper techniques and tools.										

MODULE NUMBER	COURSE CONTENT
1	Introduction to PROLOG Programming along with the IDE and its Basic Components Assignments for understanding the Basic Components of Knowledge Representation and Inferencing in Artificial Intelligence using PROLOG Programming and its working strategy.
2	Arithmetic, Boolean Expression, Decision Making Strategies Assignments for understanding implementation of Arithmetic Expression, Boolean Expression, and Decision-Making Strategies.
3	Recursion and Looping through Recursion Assignments for understanding implementation of Recursion and Looping through Recursion.
4	List of Data Items in PROLOG Assignments for understanding the utility of List in solving various problems.
5	Blind Search Techniques – BFS, DFS Implementation of BFS and DFS Algorithms for Goal Searching to solve Puzzles (8-Puzzle, Water Jug Puzzle)
6	Heuristic Search Techniques – A* Search Implementation of A* Search Algorithm for Goal Searching to solve Puzzles (8-Puzzle, Route Finding Puzzle)
7	Constraint Satisfaction Problem Solving Implementation of Backtracking Strategies to solve Constraint Satisfaction Problems (Graph Coloring Problem, 8-Queens Problem)
8	Game Playing Implementation of Adversarial Search Algorithm with alpha-beta pruning strategy for Game Playing (Tic-Tac-Toe)
9	Discussion on Project Problems and Allocation (Problem Description Report Submission)
10	Designing Solution Model and Proposal Report Submission

[illegible]

SYLLABUS	
Semester – 5 th	
Course Code	CT592
Course Name	Database Management System Lab
Lecture (per week)	0
Tutorial (per week)	0
Contact Hours (per week)	3
Total Contact Hours	36
Credit	1.5
Prerequisites: 1. Logic of programming language 2. Basic concepts of data structure and algorithms	
Course Outcome: After successful completion of this course, students will be able to:	
CO1	Demonstrate and Explain the database management system and different database languages.
CO2	Understand and Apply the SQL queries related to management of data and transaction processing for solving real life problems.
CO3	Explain and Analyze about query processing techniques involved in query optimization.
CO4	Demonstrate and Apply the PL/SQL programming, the concept of Cursor Management, Error Handling, Package and Triggers for solving real life complex problems.
CO5	Design and Assess the commercial database systems.
MODULE NUMBER	COURSE CONTENT
1	Conceptual Designing using ER Diagrams (Identifying entities, attributes, keys and relationships between entities, cardinalities, generalization, specialization etc.)
2	Converting ER Model to Relational Model (Represent entities and relationships in Tabular form, Represent attributes as columns, identifying keys) and apply the normalization techniques.
3	Creation of Tables using SQL-Overview of using SQL tool, Data types in SQL, Creating

	Tables(along with Primary and Foreign keys), Altering Tables and Dropping Tables
4	Practicing DML commands- Insert, Select, Update, Delete
5	Practicing Queries using ANY, ALL, IN, EXISTS, NOT EXISTS, UNION, INTERSECT, CONSTRAINTS etc., Practicing Sub queries (Nested, Correlated) and Joins (Inner, Outer and Equi).
6	Practice Queries using COUNT, SUM, AVG, MAX, MIN, GROUP BY, HAVING, VIEWS Creation and Dropping, Practicing on Triggers - creation of trigger, Insertion using trigger, Deletion using trigger, Updating using trigger.
7	Procedures- Creation of Stored Procedures, Execution of Procedure, and Modification of Procedure, PL/SQL, Cursors- Declaring Cursor, Opening Cursor, Fetching the data, closing the cursor.

CO-PO Mapping:

CO#	PO 1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO1 0	PO1 1	PO1 2	PSO 1	PSO 2	PSO 3
CO1	3	3	2	2	2	2	-	-	-	-	-	-	3	3	2
CO2	3	3	3	3	3	1	1	-	-	-	1	1	3	3	3
CO3	3	3	3	3	3	2	1	-	-	1	1	1	2	3	2
CO4	3	3	3	3	3	2	1	-	-	1	1	-	3	3	3
CO5	3	3	3	3	3	2	1	1	-		2	1	3	3	3

SYLLABUS											
Semester – 5 th											
Course Code	CS593										
Course Name	Object Oriented Programming using Java Lab										
Lecture (per week)	0										
Tutorial (per week)	0										
Contact Hours (per week)	3										
Total Contact Hours	36										
Credit	1.5										
Course Objectives: The objectives of this course are to enable students to • It demonstrates that how can you change the implementation of an object without affecting any other code by increasing data security and protecting unwanted data access. (Encapsulation). • It allows you to have many different functions, all with the same name, all doing the same job, but depending upon different data. (Polymorphism). • It guides you to write generic code: which will work with a range of data, so you don't have to write basic stuff over, and over again. (Generics). • It lets you write a set of functions, then expand them in different direction without changing or copying them in any way. (Inheritance)											
Course Outcome: After successful completion of this course, students will be able to: <table> <tr> <td>CO1</td><td>Create the procedure of communication between Objects, classes & methods.</td></tr> <tr> <td>CO2</td><td>Understand the elementary facts of Object Orientation with various characteristics as well as several aspects of Java.</td></tr> <tr> <td>CO3</td><td>Analyze distinct features of different string handling functions with various I/O operations.</td></tr> <tr> <td>CO4</td><td>Discuss simple Code Reusability notion w.r.t. Inheritance, Package and Interface.</td></tr> <tr> <td>CO5</td><td>Apply Exception handling, Multithreading and Applet (Web program in java) programming concept in Java.</td></tr> </table>		CO1	Create the procedure of communication between Objects, classes & methods.	CO2	Understand the elementary facts of Object Orientation with various characteristics as well as several aspects of Java.	CO3	Analyze distinct features of different string handling functions with various I/O operations.	CO4	Discuss simple Code Reusability notion w.r.t. Inheritance, Package and Interface.	CO5	Apply Exception handling, Multithreading and Applet (Web program in java) programming concept in Java.
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	4. Programming on this keyword, call by value, static variables & methods, inner classes
2	Basic String handling & I/O[12L] 1. Programming to show the use of String class methods - charAt(), compareTo(), equals(), equals Ignore Case(), indexOf(), length() , substring(), toCharArray(), to Lower Case(), toString(), to UpperCase(), trim(), valueOf() methods. 2. Programming to show the use of String Buffer class methods - append(), capacity(), charAt(), delete(), delete CharAt(), ensure Capacity(), getChars(), indexOf(), insert(), length(), setCharAt(), setLength(), substring(), toString() methods. 3. Programming on Command line arguments. 4. Programming using keyboard input by implementing Buffered Reader & Scanner classes.
3	Inheritance, Interface and Java Packages[6L] 1. Programming on Simple Inheritance, super and final keywords, super() method. 2. Programming on method overriding, dynamic method dispatch, abstract classes & methods, multiple inheritance by using interface. 3. Programming on importing system package, creating user-defined package, importing user-defined package, using protected access specifier, subclassing an imported class of a package, using same names for classes of different packages, adding multiple public classes to a package.
4	Exception handling, Multithreading and Applet Programming [8L] 1. Programming on exception handling using try-catch block, implementing throw and throws keywords, using finally block, creating user-defined exception. 2. Programming on creating child threads i) by extending thread class ii) by implementing runnable interface, creating child threads by assigning thread priorities. 3. Programming on creating simple applet to display some message, creating applet two add 2 integers, creating applet to do GUI based programming.
Textbook: 1. Herbert Schildt – "Java: The Complete Reference " – 9th Ed. – TMH 2. E. Balagurusamy – " Programming With Java: A Primer " – 3rd Ed. – TMH. Reference Books: 1. R.K Das – " Core Java for Beginners " – VIKAS PUBLISHING. 2. Rambaugh, James Michael, Blaha – " Object Oriented Modelling and Design " – Prentice Hall, India	

CO-PO Mapping:

COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	3	3	3	2	-	-	-	1	-	-	1	-	3	3
CO2	3	2	2	-	1	-	-	-	1	-	-	2	3	2	3

C03	2	3	2	3	-	-	-	-	2	-	-	-	-	3	3
C04	1	-	-	-	-	-	-	-	1	2	-	2	3	3	-
C05	2	1	1	-	1	-	-	-	2	-	-	2	-	2	3